

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. We construct genuine positive finite-volume phase components on a common sample space and prove exact reconstruction of the physical law up to a superexponentially negligible remainder. Phase-local analytic charts coexist with Lee–Yang pinching, while a microscopic subexponential mixture theorem—not a formal conjugacy of a maximum pressure—gives the minimum component rate. The labelled projective LDP, shell coefficient, local likelihood, and contraction commutation then follow.

Platform identifier: LABELLED-COMMUTE-v3. **Controlling revision:** ROUND10-REFEREE-POSITIVE-CLOSURE. **Registered proof source:** D1_MICROSCOPIC_PHASE_DISINTEGRATION.tex.

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1. ROUND-TEN RECONSTRUCTION: MICROSCOPIC PHASE DISINTEGRATION AND COMMUTING LIMITS

This paper retains a standalone theorem, but phase labels are now genuine microscopic objects. They are not external tags appended to copies of the same law.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 60F10, 82C40; Secondary 60H10, 49L25.

Key words and phrases. kinetic large deviations, tangent laws, entropic rigidity, Girsanov theorem, Boltzmann equation.

1.1. Phase-restricted finite-volume laws. For every finite projection m , let $B_{n,1}, \dots, B_{n,J_m}$ be disjoint measurable phase basins determined by an order parameter in that projection. Assume their uncovered probability is superexponentially small and their basin boundaries have subexponential probability under all compact real source tilts. Define

$$w_{n,j} = \mathbb{P}_n(B_{n,j}), \quad \mathbb{P}_{n,j} = \mathbb{P}_n(\cdot \mid B_{n,j}), \quad Z_{n,j}(\theta) = \mathbb{E}_n[e^{n\theta \cdot X_n}; B_{n,j}].$$

The platform papers construct these basins from their exposed phase order parameters and prove the boundary estimates.

Theorem 1.1 (Microscopic phase mixture LDP). *Suppose $n^{-1} \log w_{n,j} \rightarrow -\alpha_j$ and every restricted law $\mathbb{P}_{n,j}$ satisfies a good LDP with rate I_j . Then the original unlabelled law satisfies a good LDP with*

$$I(x) = \min_j \{\alpha_j + I_j(x)\}.$$

The joint labelled law has rate $\alpha_j + I_j(x)$. This conclusion follows from the microscopic decomposition, not from the false identity $(\max_j Q_j)^ = \min_j Q_j^*$.*

Proof. For a closed set, sum the finitely many phase probabilities and use the largest exponential term; the superexponential remainder is irrelevant. For an open set, choose one phase attaining the minimum and use its lower bound together with the phase weight. Exponential tightness of the restricted laws and finiteness of the phase set give goodness. Keeping the label makes the same proof immediate on the disjoint union. $\square$

1.2. Phase-local complex charts.

Theorem 1.2 (Phase-local response and Lee–Yang-compatible coexistence). *For every phase j and compact subset of its real regularity domain,*

$$Q_{n,j}(\theta) = n^{-1} \log Z_{n,j}(\theta)$$

has a uniform zero-free complex neighborhood and converges there with all derivatives to Q_j . No common zero-free tube is asserted across a coexistence surface. The physical pressure is

$$Q(\theta) = \max_j \{Q_j(\theta) - \alpha_j\},$$

and Lee–Yang zeros may approach its nondifferentiable phase boundary.

Proof. Inside a phase basin the platform transfer/cluster operator has a simple isolated eigenvalue and the basin-boundary remainder is subexponential. A uniform Riesz contour gives the restricted zero-free logarithm and analytic convergence. At coexistence, two restricted exponentials have comparable real parts and their sum can have zeros at distance $O(n^{-1})$; the theorem makes no contrary assertion. Laplace’s principle over the finite phase mixture gives the maximum formula. $\square$

Corollary 1.3 (Exact finite-centred local likelihood). *Within a regular phase, for a finite source direction h ,*

$$L_{n,j}(h) = \exp \left\{ \sqrt{n} h \cdot (X_n - DQ_{n,j}(0)) - n[Q_{n,j}(h/\sqrt{n}) - Q_{n,j}(0) - n^{-1/2} DQ_{n,j}(0)h] \right\}$$

is an exact mean-one density and converges to the Cameron–Martin likelihood with covariance $D^2Q_j(0)$.

Proof. The normalization is the finite restricted log moment generating function, so expectation is exactly one. Taylor expansion on the phase-local complex chart and the finite-dimensional Gaussian limit give convergence and uniform integrability. $\square$

1.3. Faces and projective passage.

Theorem 1.4 (Labelled projective LDP and topology upgrade). *If every finite observable projection has the phase-labelled LDP above and the platform laws are exponentially tight in the weighted path topology, then the joint labelled path law satisfies a full good LDP with the supremum of the consistent finite rates. Every finite-rate boundary-face point has a recovery sequence obtained inside one restricted phase basin or by the microscopic phase mixture.*

Proof. Exact contraction of each restricted law gives projective consistency. Dawson–Gärtner yields the projective LDP and exponential tightness upgrades it to the weighted topology. For a target-open set, choose a compact rate sublevel and a finite projection whose inverse image separates the target point inside that compact set. Use the phase-specific recovery law; its exponential tightness makes probability outside the compact negligible under the same tilt. Boundary faces are obtained by the platform’s restricted normal-source limit before the projective limit. No Straszewicz statement is substituted for microscopic recovery. $\square$

1.4. Thin shells through phase mixtures.

Theorem 1.5 (Exact labelled and unlabelled shell conditioning). *Suppose the A2 or B1 local coefficient holds uniformly on every active phase chart. Conditioning a restricted phase law on a regular thin shell gives its source-dependent information projection. Conditioning the original mixture gives the finite mixture of active projected phases with weights equal to the phase weights times their shell coefficients. The logarithmic conditioning cost is the minimum of the phase costs.*

Proof. Apply the phase-local coefficient to numerator and denominator within each basin. Sum over the finitely many phases. Exponentially nonminimal terms disappear; tied phases retain their explicit subexponential coefficients. The basin-boundary remainder is superexponential. This is an actual microscopic conditional decomposition, not a formal label appended after the limit. $\square$

1.5. The commutation theorem.

Theorem 1.6 (Phase-labelled commutation principle). *On the platform domains proved in A2–A4 and B1–B4, the following operations commute within each regular restricted phase and then with the finite microscopic mixture:*

- (i) *finite-source differentiation and the microscopic limit;*
- (ii) *source-dependent thin-shell conditioning;*
- (iii) *continuous typed contraction;*
- (iv) *formation of the exact-centered Gaussian tangent and local likelihood;*
- (v) *dynamic action concatenation and the nonlinear semigroup.*

At coexistence, derivatives are phase-wise and the physical tangent is the compact convex set or mixture selected by the microscopic phase weights.

Proof. Within one phase, uniform complex convergence gives derivative commutation; A2/B1 give the shell coefficient; ordinary and extended contraction give the observable rates; A4/B3 give Gaussian process convergence; and B4 gives the additive dynamic semigroup. The finite phase decomposition of Theorem 1.1 allows each identity to be summed before or after the limit. At coexistence no common analytic derivative is asserted; the labelled statements and explicit mixture weights give the correct set-valued or mixed result. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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